

< Previous

 ✓

 ✓

 ✓





Next >

Notifications 

4. Column space

 [Bookmark this page](#)

Upgrade your course today

- ✓ You are taking the free version of this course.
- ✓ Upgrade today and, upon passing, receive your digital certificate signed by MIT faculty to highlight the knowledge and skills you’ve gained from this MITx course.

Upgrade

 Calculator

https://courses.mitxonline.mit.edu/learn/course/course-v1:MITxT+18.03.3x+1T2024/block-v1:MITxT+18.03.3x+1T2024+type@sequential+block@recitation3_for_edx/block-v1:MITxT+18.03.3x+1T2024+type@vertical...

1/4

Recitation due Feb 14, 2024 20:30 IST

Find y

1/1 point (graded)

Find the value of y such that the columns of $\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 3 & y & 4 \\ 5 & 1 & 6 \end{pmatrix}$ linearly dependent?

$y =$ ✓ Answer: 1

Solution:

There are many ways to solve this problem.

One way to find the value of y such that the columns are the matrix of $\mathbf{A}$ are linearly dependent is to determine what value of y allows you to express the second column is a linear combination of the first and third column. Observe that

$$\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Therefore the value of y that makes the matrix singular is $y = 1$.

Another solution is to compute the determinant in terms of y , and find the value of y that makes the determinant 0:

$$\begin{aligned} \det(\mathbf{A}) &= 1(6y - 4) - 1(18 - 20) + 2(3 - 5y) \\ &= -4y + 4. \end{aligned}$$

Therefore the determinant equals zero when $y = 1$.

Submit

You have used 1 of 5 attempts

i Answers are displayed within the problem

A vector not in the column space

1/1 point (graded)

Let $\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 3 & y & 4 \\ 5 & 1 & 6 \end{pmatrix}$ be the matrix where y is the value you found in the previous problem.

Which of the vectors are **not** in the column space of $\mathbf{A}$? (Choose all that are not in the column space.)

- ☐ $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$
- ☐ $\begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}$

☒ $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$

☒ $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$

☒ $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$



Solution:

We will go through the columns one by one.

- $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is the second column of $\mathbf{A}$, therefore it is in the column space of $\mathbf{A}$.

- $\begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}$ is the first column of $\mathbf{A}$, therefore it is in the column space of $\mathbf{A}$.

- $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ is perpendicular to both of the columns of $\mathbf{A}$. If we dot this vector with the two column vectors we get zero. Therefore this vector is orthogonal to every vector in the span of the columns, it is not in the column space. In fact, we can use this vector to give a defining equation for the column space. The column space of $\mathbf{A}$ is the same as the nullspace of $\begin{pmatrix} 1 & -2 & 1 \end{pmatrix}$, in other words, all solutions to the equation $x - 2y + z = 0$.

- $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ is not in the column space because it does not satisfy the defining equation.

- $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ is not in the column space because it does not satisfy the defining equation.

You have used 1 of 4 attempts

i Answers are displayed within the problem

MATLAB exploration

Continue to set $\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 3 & y & 4 \\ 5 & 1 & 6 \end{pmatrix}$, where the value of y is such that the columns are linearly dependent as found in the first problem.

Calculator

1/14/24, 10:39 PM

MATLAB Recitation 3 | Unit 2: Linear Algebra, Part 2 | Linear Algebra and NxN Systems | MITx Online

Let the vector **c** be a vector not in the column space of **A**. (See the previous problem.)

PreviousNext

Then **Ax = c** has no solution. What solution do you find if you try to solve this equation using MATLAB?

Let the vector **b** be the first column of **A**. What solution do you find if you type A

© All Rights Reserved

© MITx Online. All rights reserved except where noted.

[About Us](#)

[Terms of Service](#)

[Privacy Policy](#)

[Honor Code](#)

[Accessibility](#)

 Calculator